The DATA-MD Curriculum

Session 3 — Automation and Benchmarks

Data-Driven, Technology-Assisted Medical Decision Making for Medical Trainees

Balczewski · Wong · James · Goparaju · East · Baker · Nguyen · Wang · Zickuhr



Session 3: Learning Objectives

Levels of Automation

Understand general paradigms for **human-AI teams** and where clinicians fit in the loop.

Critical Appraisal

Evaluate a study **benchmarking AI** for clinical decision making — rigorously and skeptically

The Noise Around AI Clinical Reasoning

Real study titles + media headlines

● Studies ● Media headlines

Science

Performance of a Large Language Model on the Reasoning Tasks of a Physician

Science, 2026

NEJM AI

Use of GPT-4 to Diagnose Complex Clinical Cases

NEJM AI, 2024

NEJM AI

GPT versus Resident Physicians — A Benchmark Based on Official Board Scores

NEJM AI, 2024

JAMA Network Open

Large Language Model Influence on Diagnostic Reasoning: A Randomized Clinical Trial

JAMA Network Open, 2024

nature

Towards accurate differential diagnosis with large language models

Nature, 2025

JAMA

Accuracy of a Generative Artificial Intelligence Model in a Complex Diagnostic Challenge

JAMA, 2023

JAMA Network Open

Large Language Model Performance and Clinical Reasoning in Multistep Clinical Vignettes

JAMA Network Open, 2026

nature medicine

Toward expert-level medical question answering with large language models

Nature Medicine, 2025

NPR

In real-world test, an AI model did better than ER doctors at diagnosing patients

NPR, Apr 30 2026

The Guardian

AI outperforms doctors in Harvard trial of emergency triage diagnoses

The Guardian, 2026

HARVARD MAGAZINE

AI Outperforms Doctors in Emergency Room Tasks, New Harvard Study Shows

Harvard Magazine, 2026

CNET

AI Outperforms ER Doctors in Diagnostic Cases, Study Points to Collaborative Care

CNET, 2026

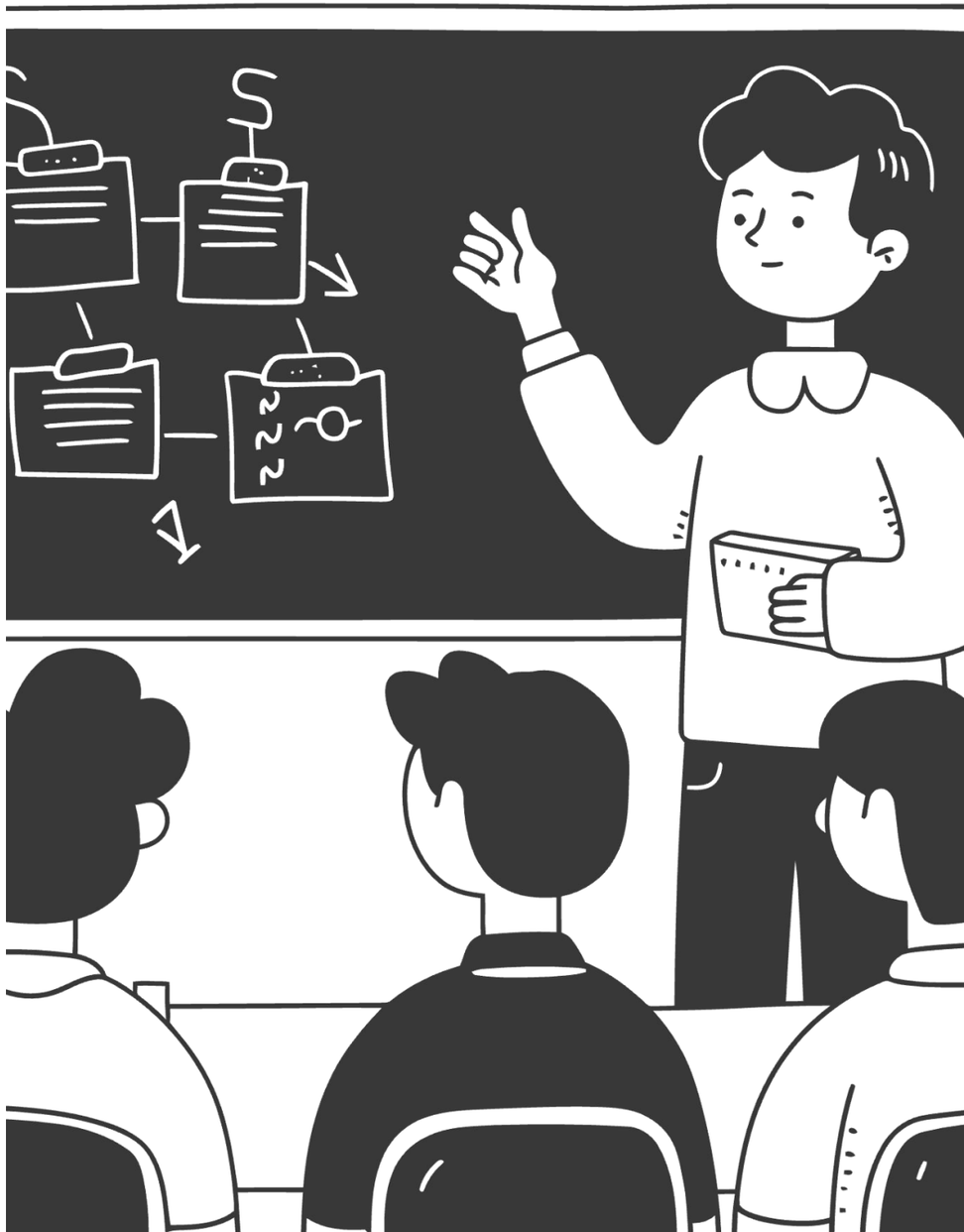
Big claims. Different benchmarks. Read the methods.

Headline excitement does not automatically equal robust clinical reasoning evaluation.

https://www.youcanknowthings.com/did-ai-really-beat-er-doctors-at-er-triage/

Session 2 - Review

Before diving into automation, let's revisit key concepts from last session.



Common Generative AI Sources of Error

1: Hallucination

Fictional, erroneous, or unsubstantiated information presented as fact

a) Inaccuracy: Correct structure, but one or more details are factually wrong

b) Misattribution: Wrong source cited, or source misinterpreted

2: Omission

Clinically important information missing from the output

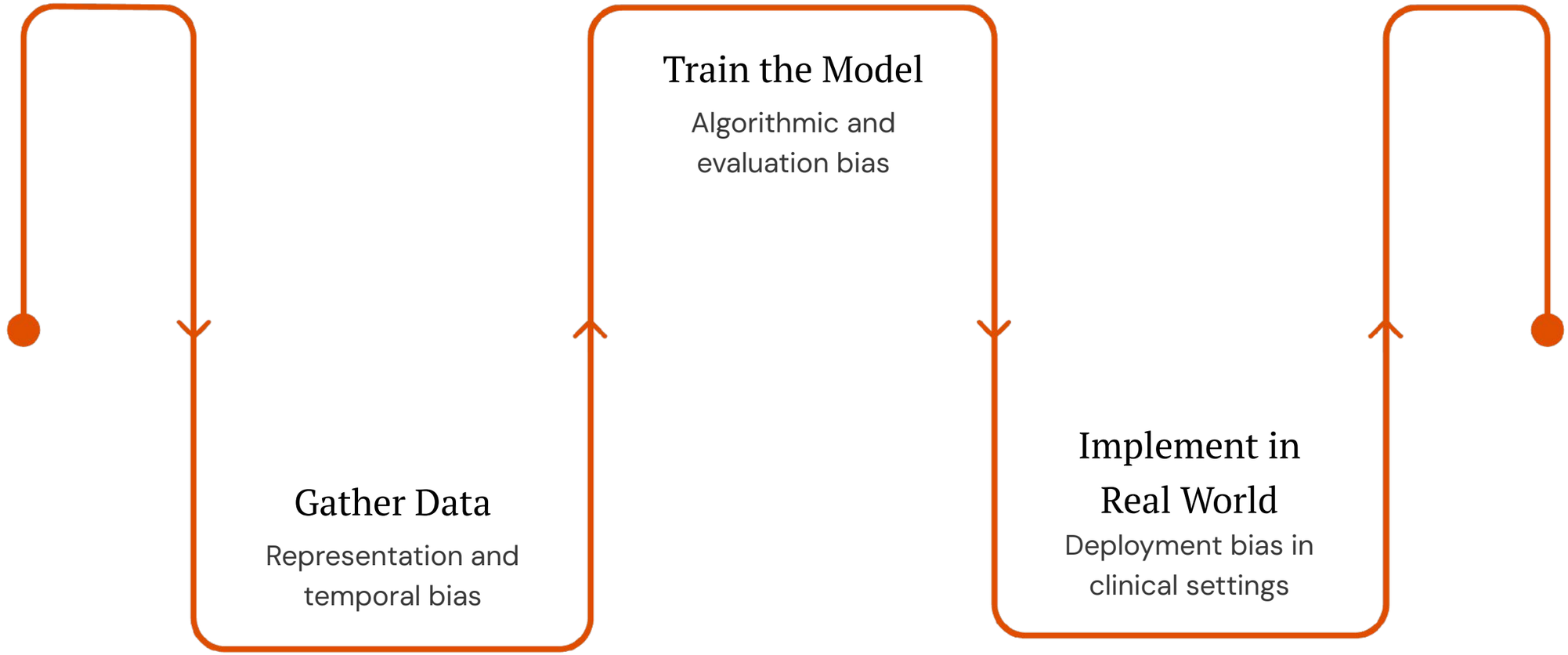
3: Limited Reasoning

Draws incorrect conclusions even from correct premises

4: Text Output Errors

Spelling, translation, grammar, or fluency mistakes

Bias Across the AI Model Lifecycle



Bias can enter at **any stage** — from who is represented in training data to how the model behaves once deployed in a clinical setting.

Generating a Good Prompt

01

Be Clear and Specific

02

Frame Your Context

03

Experiment with Iterative Questions

04

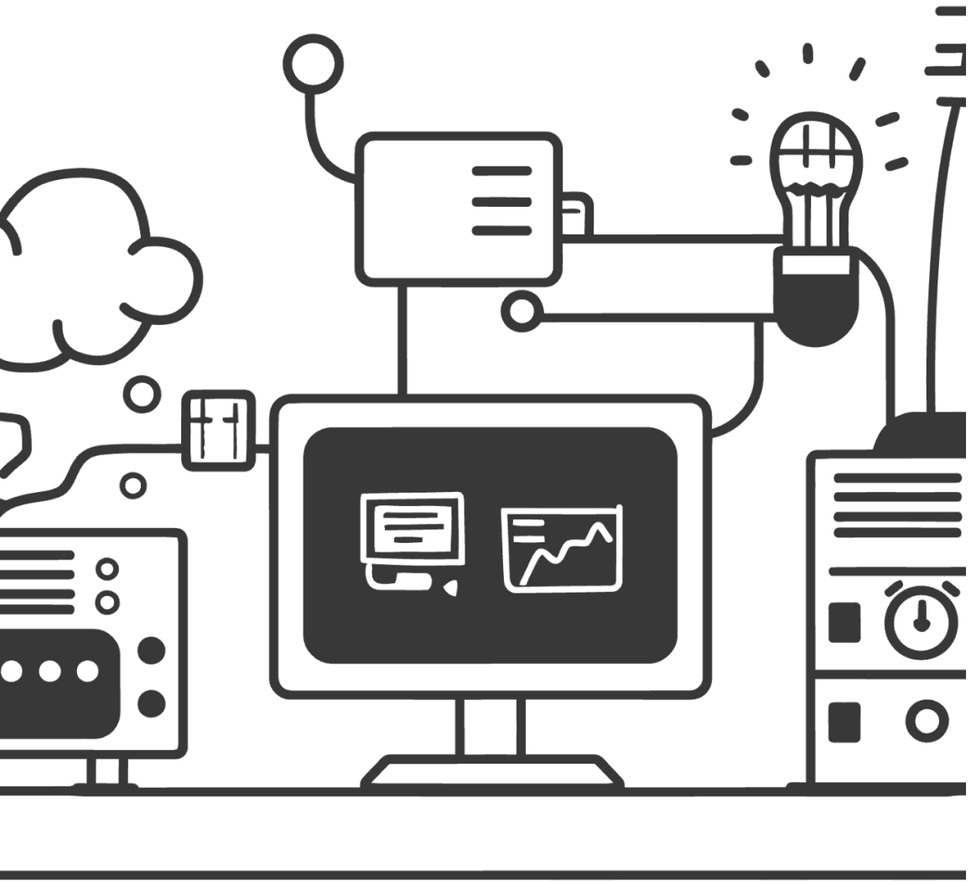
Limit Potential Bias

05

Analyze and Adapt

More techniques at **learnprompting.com**





Advanced Prompt Engineering Techniques

Meta-Prompting

Ask AI to help you write a better prompt — "prompt about the prompt"

Self-Criticism Prompting

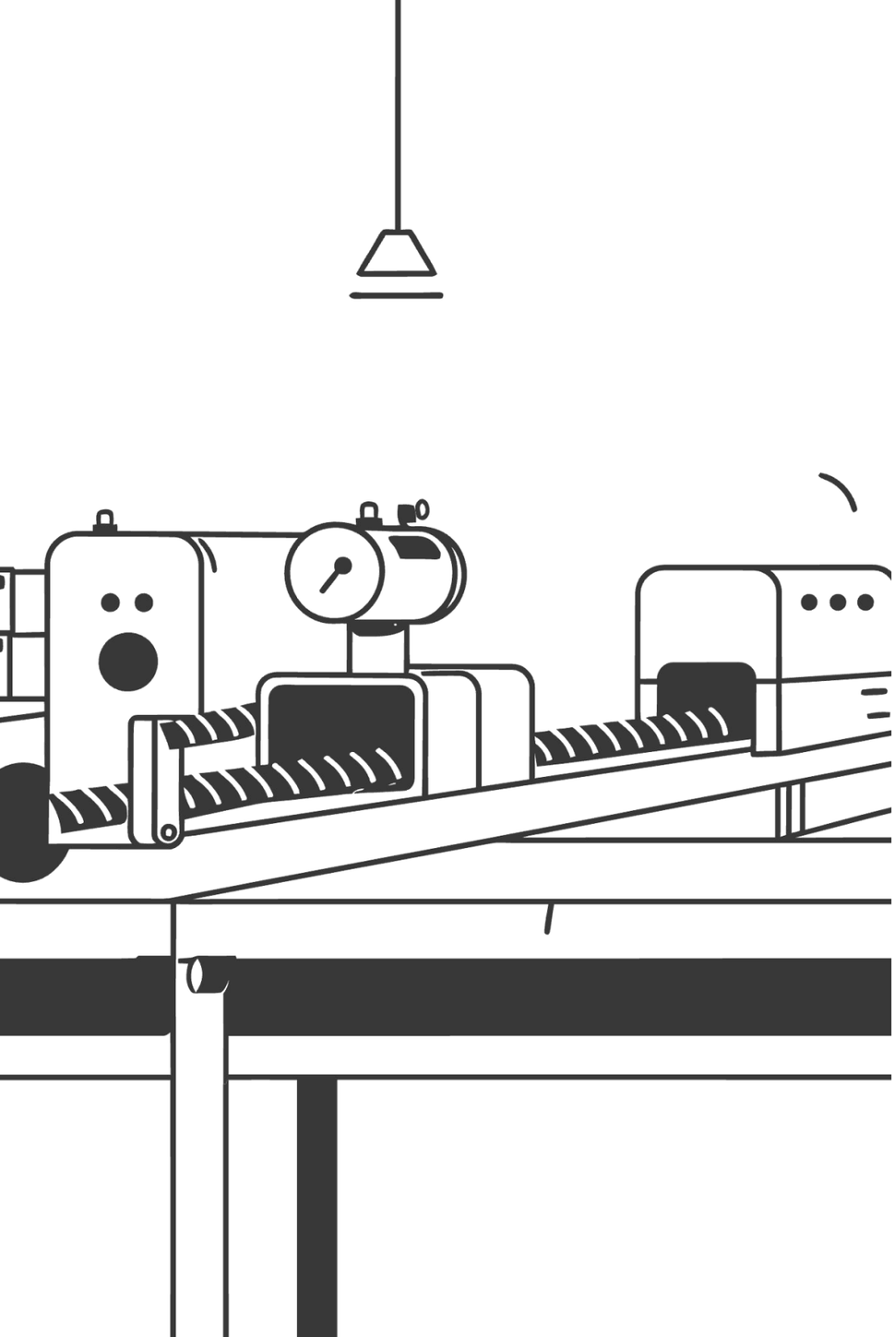
Instruct the model to critique and revise its own output

Chain-of-Thought Prompting

Guide the model to reason step by step before reaching a conclusion

Brevity

Concise, focused prompts reduce noise and improve output quality

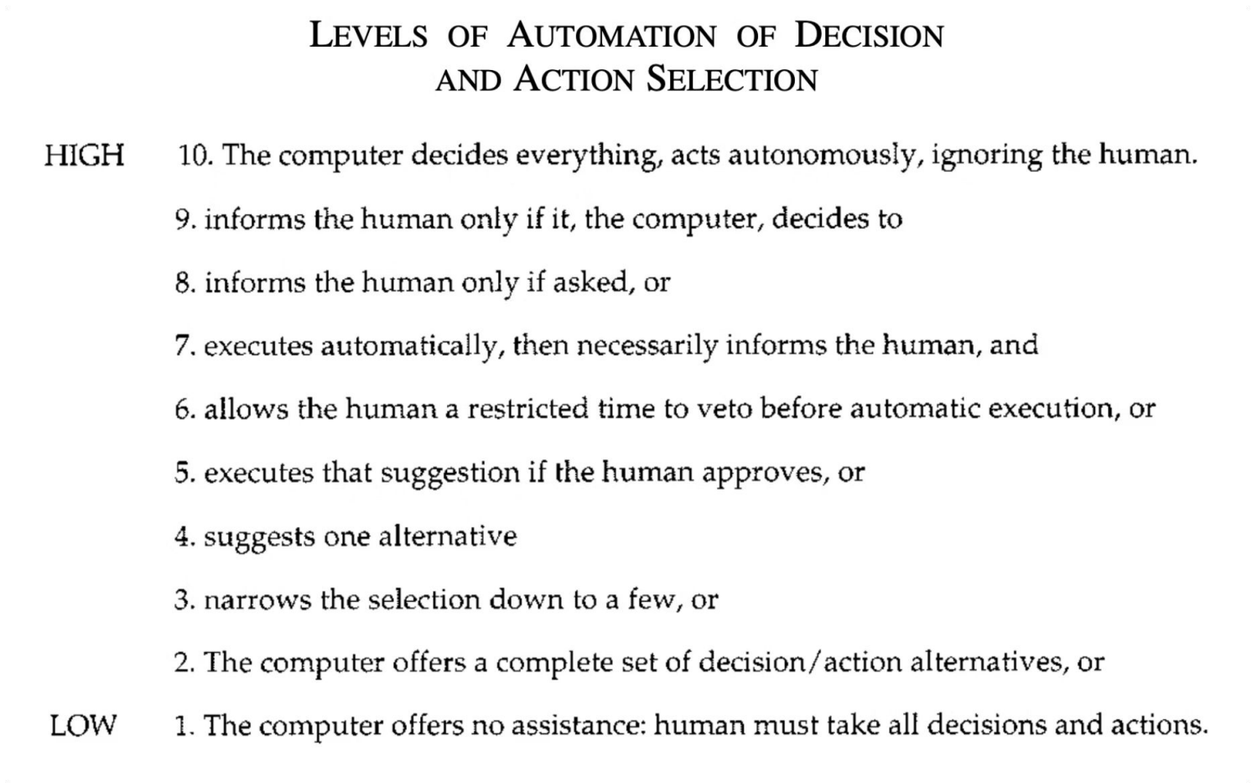


NEW TOPIC

Levels of Automation

How much control does the AI have — and how much do *you* retain?

10 Levels of Automation for Decision Making



LEVELS OF AUTOMATION OF DECISION AND ACTION SELECTION

HIGH	10. The computer decides everything, acts autonomously, ignoring the human.
	9. informs the human only if it, the computer, decides to
	8. informs the human only if asked, or
	7. executes automatically, then necessarily informs the human, and
	6. allows the human a restricted time to veto before automatic execution, or
	5. executes that suggestion if the human approves, or
	4. suggests one alternative
	3. narrows the selection down to a few, or
	2. The computer offers a complete set of decision/action alternatives, or
LOW	1. The computer offers no assistance: human must take all decisions and actions.

Level 10 (High)

Computer decides everything, acts autonomously — ignores human input

Levels 4–9

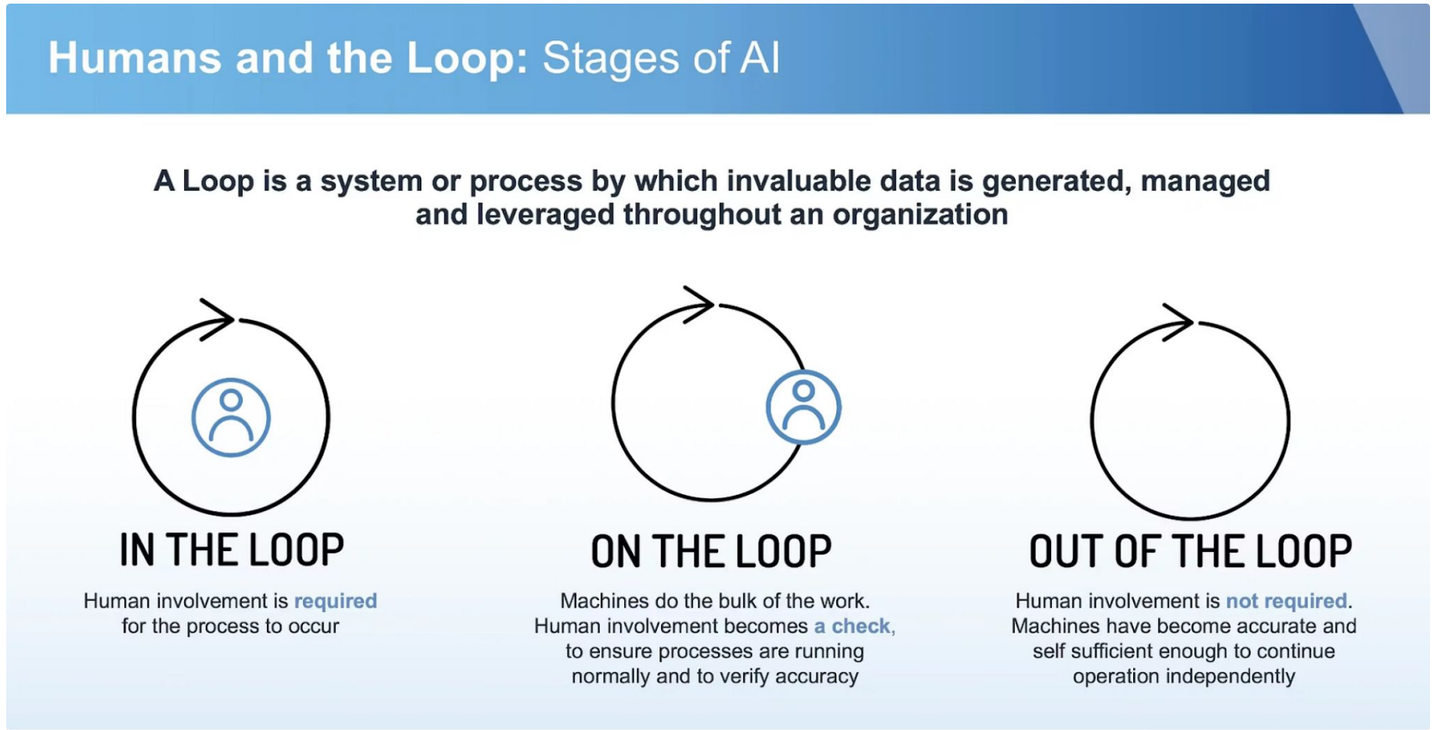
A spectrum from "suggests one alternative" to "informs human only if asked"

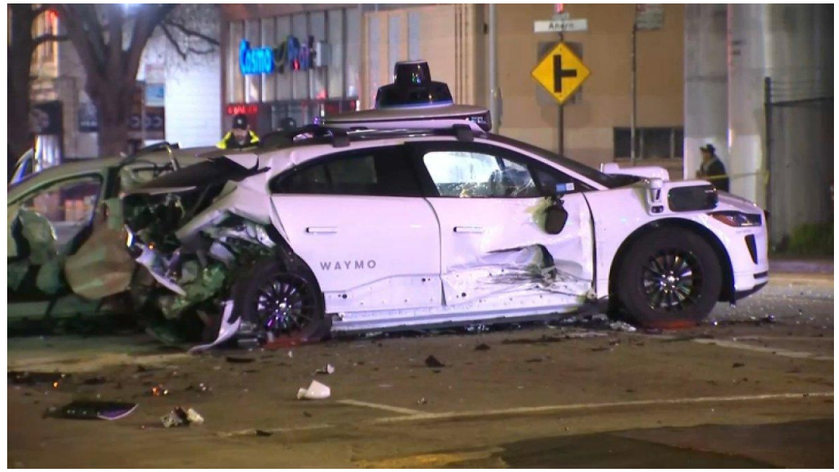
Level 1 (Low)

No AI assistance — human takes all decisions and actions

Parasuraman, IEEE Transactions on Systems, Man, and Cybernetics (2000)

Humans and the Loop





In the Loop

Human involvement is **required** for the process to occur

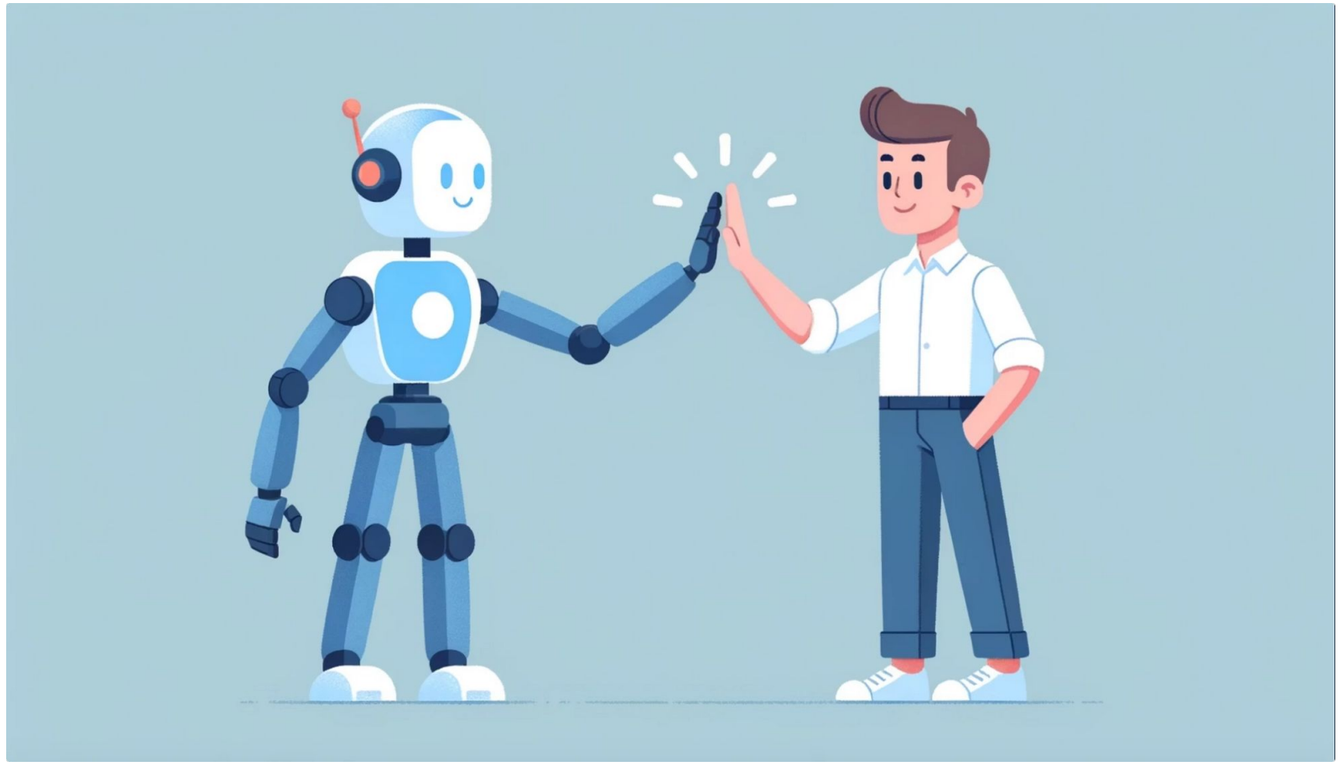
On the Loop

Machines do the bulk of work; humans **check and verify**

Out of the Loop

Human involvement is **not required** — machines operate independently

Parasuraman, IEEE Transactions on Systems, Man, and Cybernetics (2000)



Human + AI Teaming

Augmented Intelligence — the clinician and AI each contribute their strengths, with the human retaining final judgment.

AI Strengths

Pattern recognition, data processing, consistency at scale

Human Strengths

Contextual reasoning, patient values, ethical judgment

Together

Better outcomes than either alone — when the team is well-designed

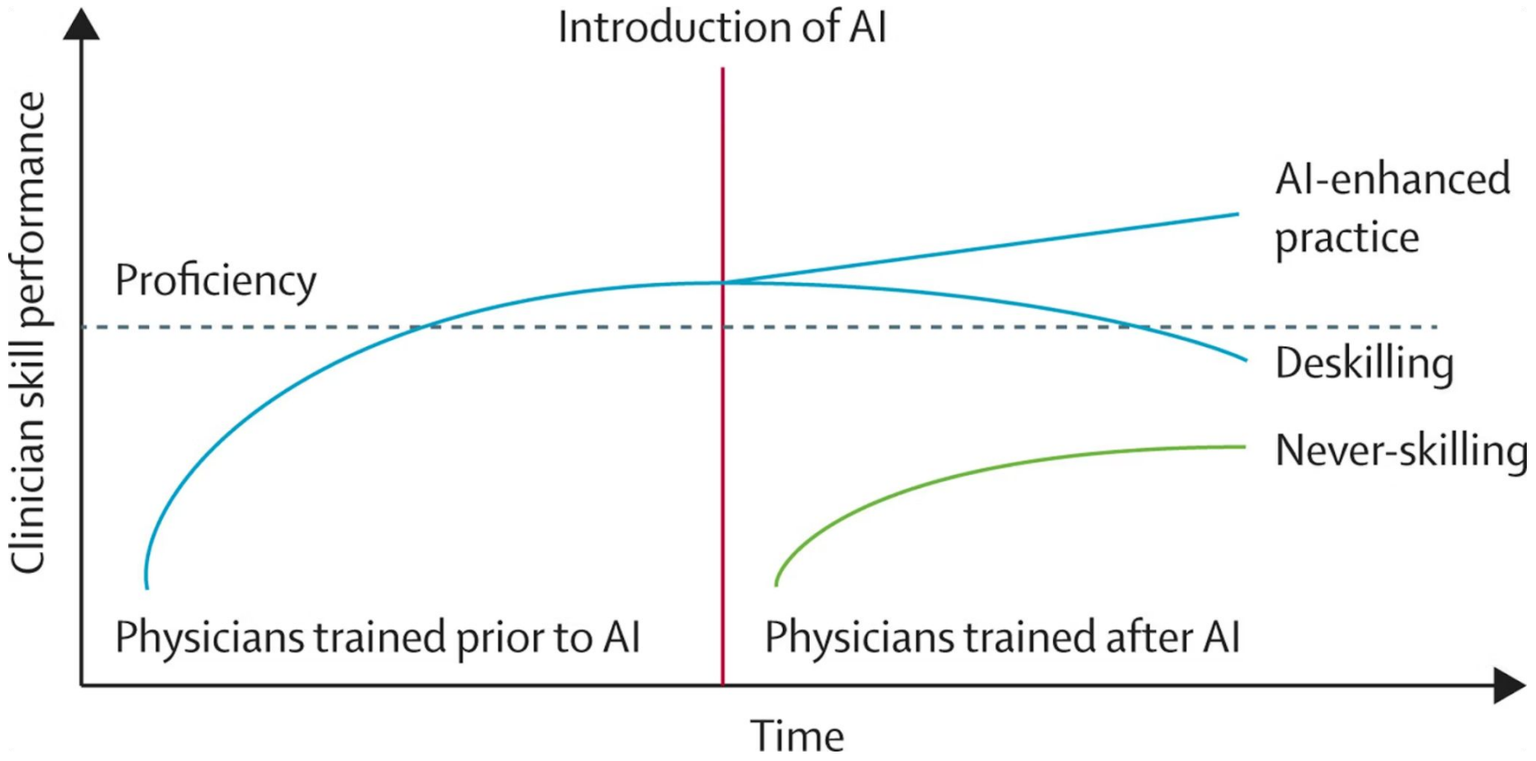


DISCUSSION

Discussion Question

What level(s) of automation have you seen among AI tools at your hospital?

Does the level of automation impact how **critically** you evaluate AI tool performance?



Drawback of Automation: Deskilling and Never-Skilling

Deskilling

Clinicians who once had skills gradually lose them when AI handles the task

Never-Skilling

Trainees who learn with AI from the start may never develop foundational competencies independently

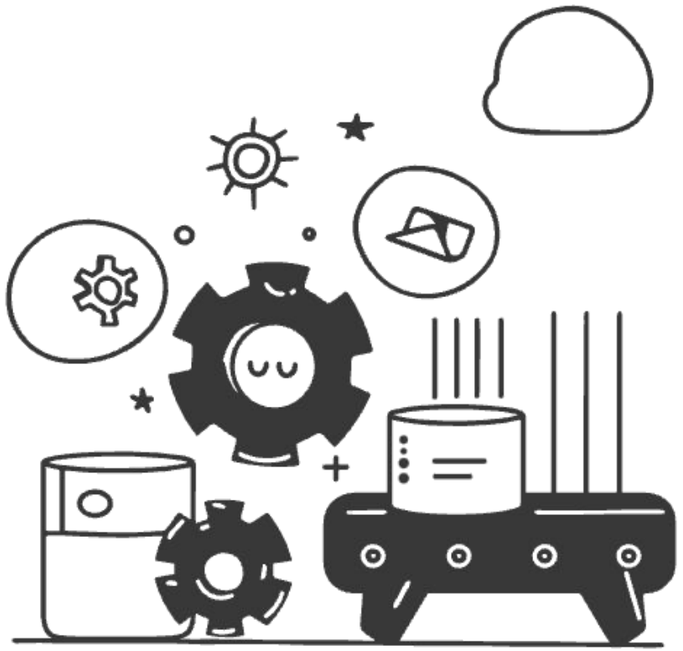
Source: The Lancet, 2025

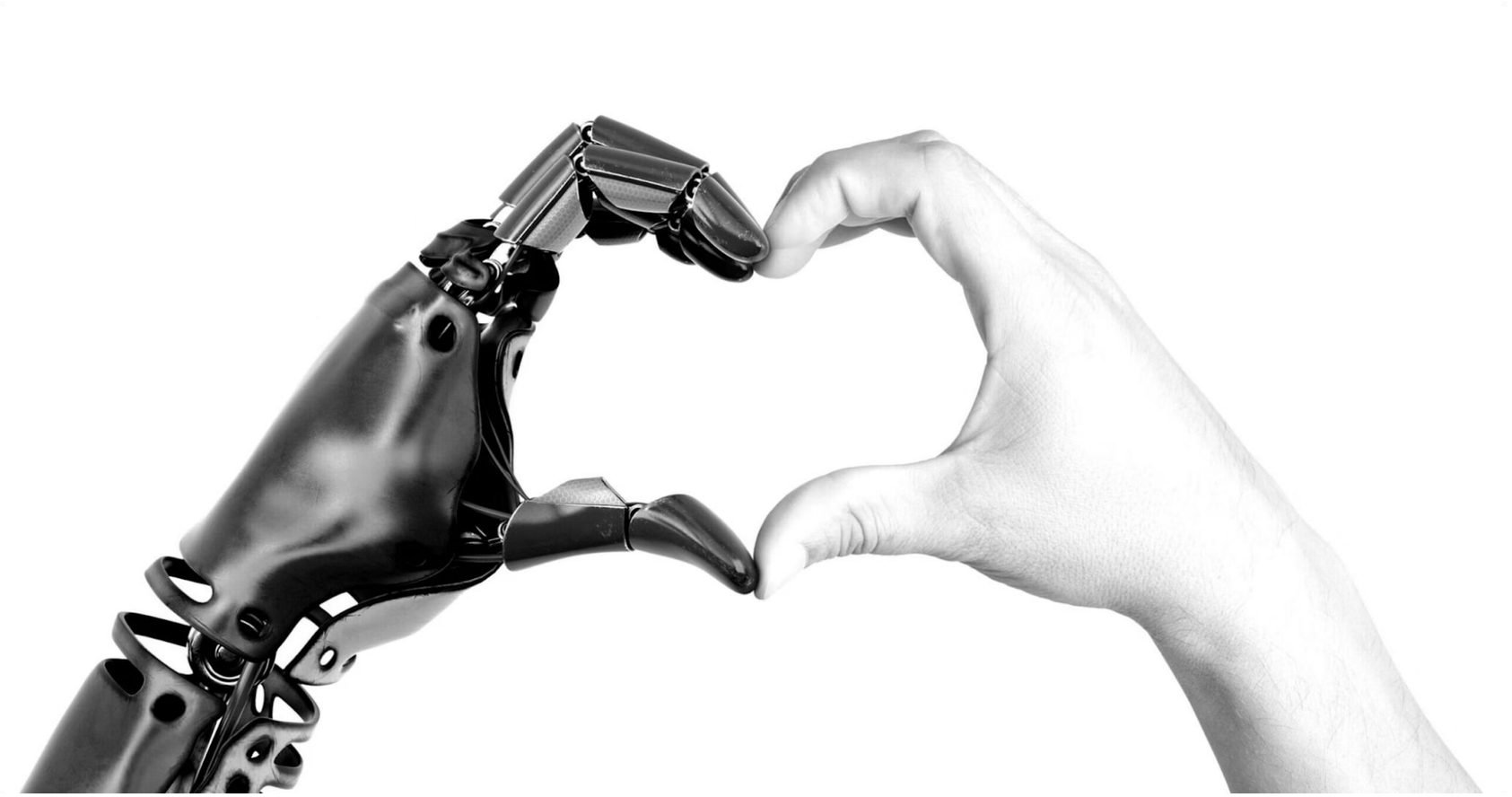
KEY CONCEPT

Reduced cognitive effort

leads to **shallower processing** and less skill development or maintenance

⚠ The more AI does, the less the clinician practices — and the more they depend on AI being correct.





NEW TOPIC

Human-AI Teaming on Clinical Decision Making

How does AI actually perform when paired with - or replacing - clinical judgment?

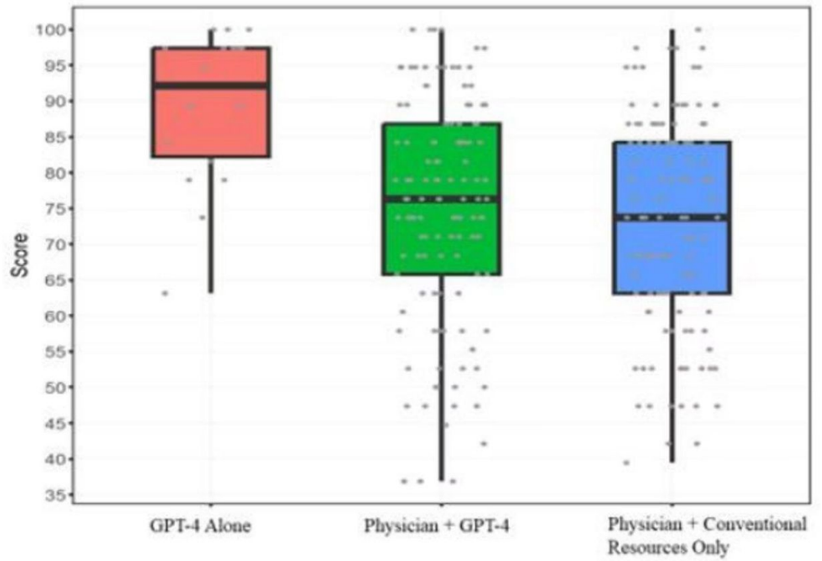
JAMA Network | Open

Original Investigation | Health Informatics

Large Language Model Influence on Diagnostic Reasoning
A Randomized Clinical Trial

Ethan Goh, MBBS, MS; Robert Gallo, MD; Jason Hom, MD; Eric Strong, MD; Yingjie Weng, MHS; Hannah Kerman, MD; Joséphine A. Cool, MD; Zahir Kanjee, MD, MPH; Andrew S. Parsons, MD, MPH; Neera Ahuja, MD; Eric Horvitz, MD, PhD; Daniel Yang, MD; Arnold Milstein, MD; Andrew P. J. Olson, MD; Adam Rodman, MD, MPH; Jonathan H. Chen, MD, PhD

eFigure 1. Distribution of Diagnostic Performance Scores of Physician + GPT-4 vs. Physician + Conventional Resources Only



AI Literature:

Benchmarking AI performance on Clinical decision making

How does the model compare to physicians?

- Standardized clinical vignettes used as test cases
- AI and clinician responses evaluated against gold-standard diagnosis

Article | Open access | Published: 04 July 2024

Evaluation and mitigation of the limitations of large language models in clinical decision-making

Paul Hager ✉, Friederike Jungmann, Robbie Holland, Kunal Bhagat, Inga Hubrecht, Manuel Knauer, Jakob Vielhauer, Marcus Makowski, Rickmer Braren, Georgios Kaissis & Daniel Rueckert

Nature Medicine **30**, 2613–2622 (2024) | Cite this article

Study results show performance varies significantly depending on how much clinical context is provided in the prompt — and who (or what) is asking the question.

AI Performs Closer to Humans with Robust Prompts

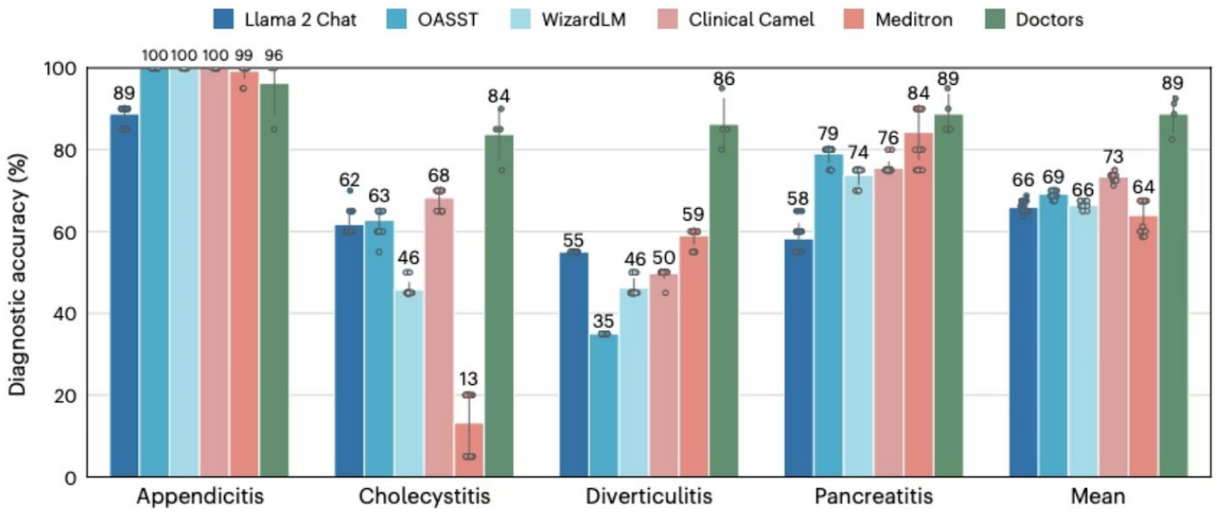
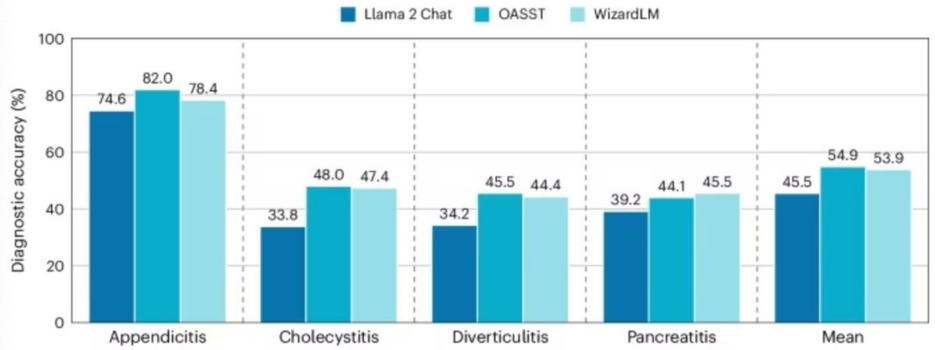


Fig. 2 | LLMs diagnose significantly worse than doctors when provided with all information. On a subset ($n = 80$) of MIMIC-CDM-FI, we compared the mean diagnostic accuracy of LLMs over multiple seeds ($n = 20$) with clinicians ($n = 4$) and found that LLMs perform significantly worse on average ($P < 0.001$) and especially on cholecystitis ($P < 0.001$) and diverticulitis ($P < 0.001$). The mean diagnostic accuracy is shown above each bar. Vertical lines indicate the standard deviation. The individual data points are shown.

When given **complete patient information**, AI diagnostic accuracy approaches physician-level performance.

…But Much Worse When Gathering Information Itself

Fig. 3: Diagnostic accuracy of LLMs decreased in an autonomous clinical decision-making scenario.



When tasked with gathering all information required for clinical decision-making themselves, LLMs perform best when diagnosing appendicitis but perform poorly on the other three pathologies of cholecystitis, diverticulitis and pancreatitis. In such a realistic clinical scenario, model performance decreased compared to the retrospective diagnosis with all information provided (MIMIC-CDM-FI). The exact diagnostic accuracy is shown above each bar.

⊗ Performance **drops sharply** when AI must independently gather clinical information — a critical limitation for real-world deployment.

Common AI Failures in Clinical Reasoning

Missing Key Data

Forgot to ask for physical exam findings, pancreatic enzymes, or abdominal imaging

Misinterpreted Results

Incorrectly interpreted lab or imaging findings, leading to flawed reasoning

Guideline Non-Adherence

Failed to follow diagnostic or management guidelines — especially in emergent cases like **perforated colon due to diverticulitis**

⚠ These failures reinforce why **human oversight** remains essential — AI is not yet safe to operate out of the loop for complex clinical decisions.

Background for Critical Appraisal

Foundations of AI literacy for evaluating clinical AI literature

AI Models & Key Players

Anthropic
Claude

OpenAI
GPT

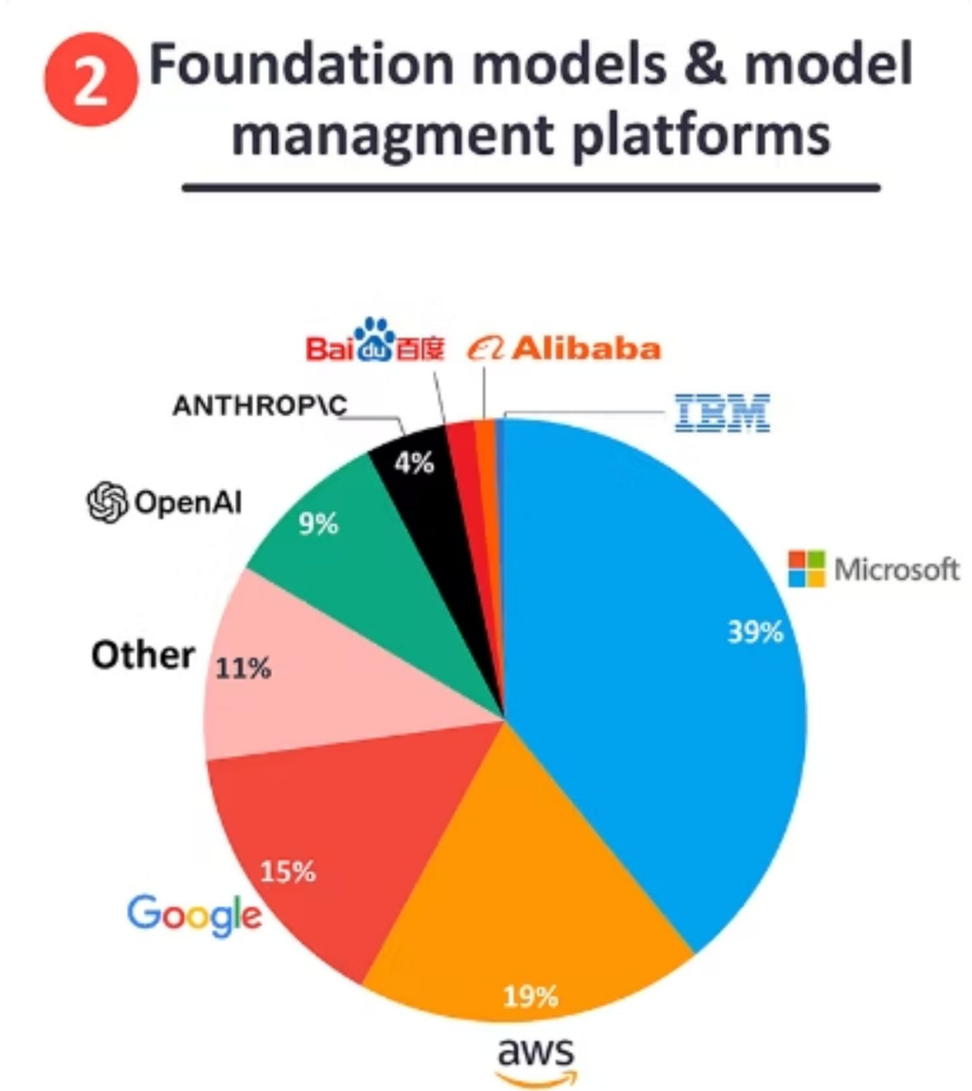
Google
Gemini

xAI
Grok

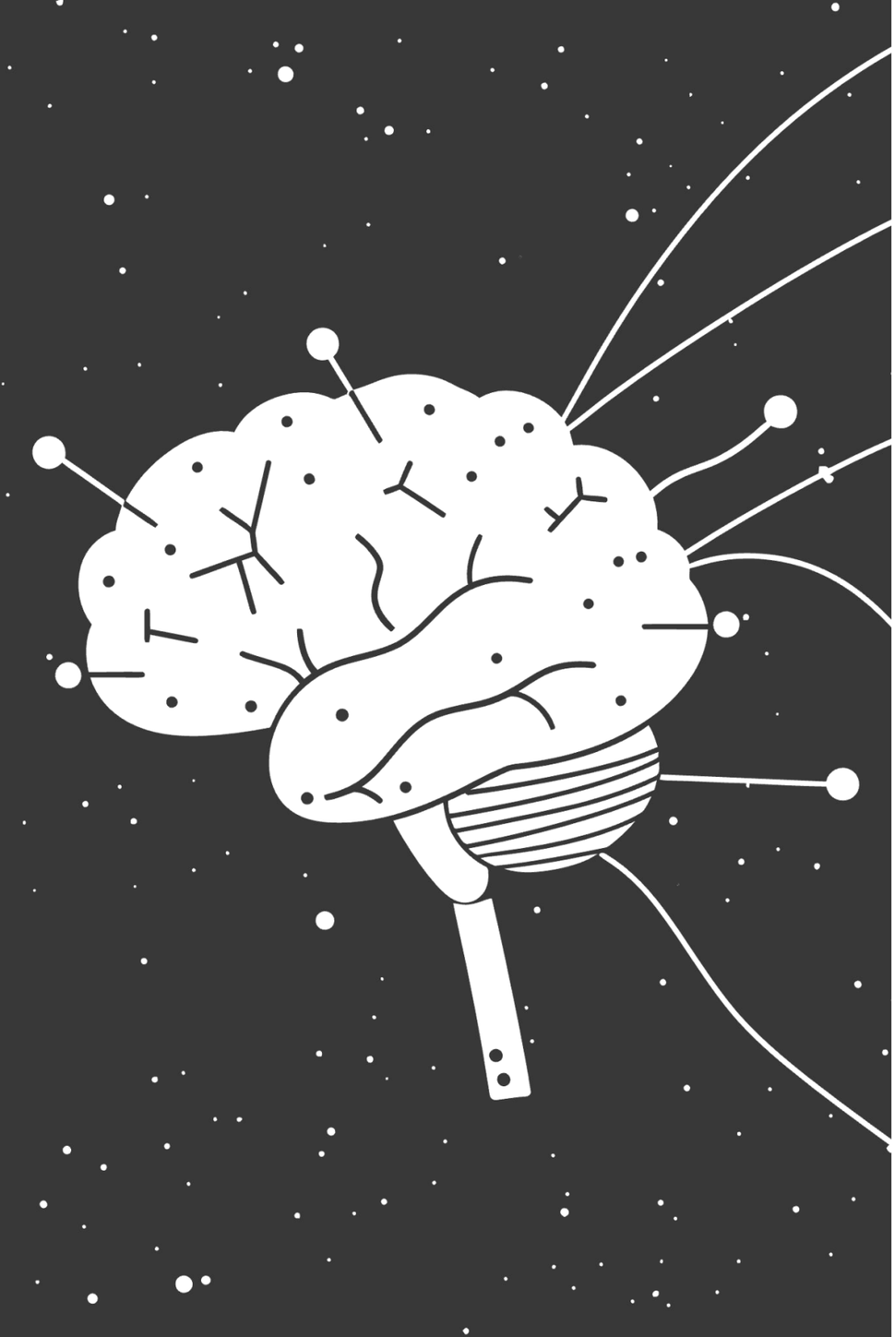
Meta
Llama

DeepSeek
DeepSeek

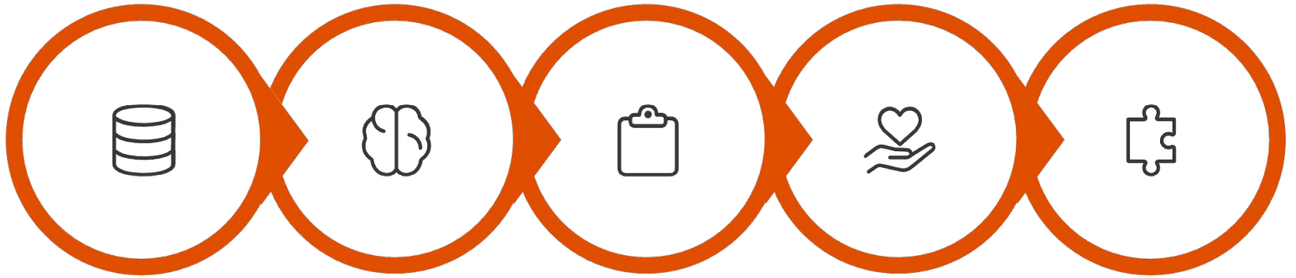
Mistral
Mistral



Microsoft leads foundation model platforms at 39% market share. Source: IoT Analytics



LLM Flavors: Foundation vs. Reasoning vs. Agent



Reasoning models: Given extra compute/time to think; use chain-of-thought reasoning to work through complex problems step by step. **Examples: Open AI o3, o4-mini, GPT-5 thinking.**

LLM Flavors: Foundation vs. Reasoning vs. Agent

Training Pipeline

01

Gather Data

Real and synthetic sources

02

Pretraining

Self-supervised learning on language/image tasks

03

Fine-tuning & RLHF

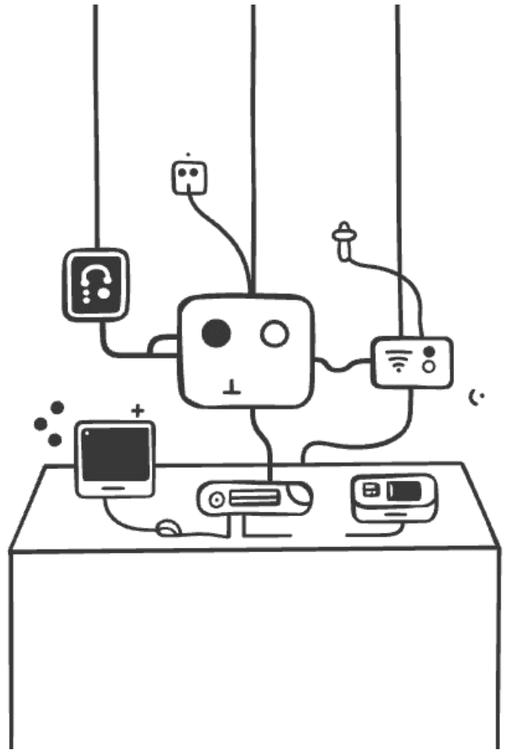
Task-specific examples + human preference scoring

04

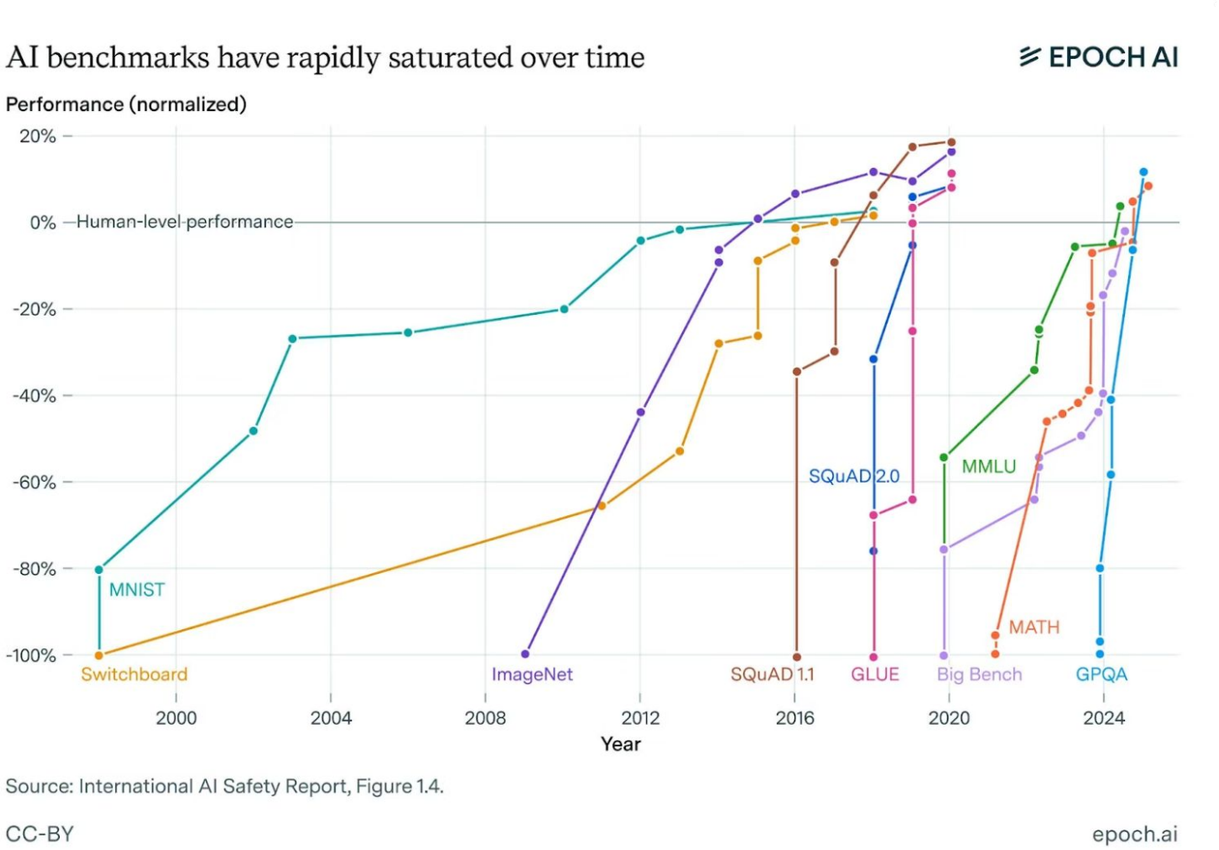
Deploy & Monitor

Continuous oversight cycles

Agentic models: Able to access the internet and external devices through application programming interfaces (APIs): enabling autonomous, multi-step task completion beyond simple text generation.



Benchmarking



Source: International AI Safety Report, Figure 1.4.

CC-BY

epoch.ai

Key Benchmarks

MNIST

Identify 70,000 handwritten digits 0–9

SQuAD

Reading comprehension on Wikipedia passages

MATH

Pre-algebra through calculus problem sets

MMLU / GPQA

Multidomain knowledge and expert-level Q&A

⚠ AI benchmarks saturate rapidly: models reach human-level performance, forcing constant benchmark renewal.



Efficacy vs. Effectiveness

Efficacy

Does it work under **ideal, controlled** conditions?

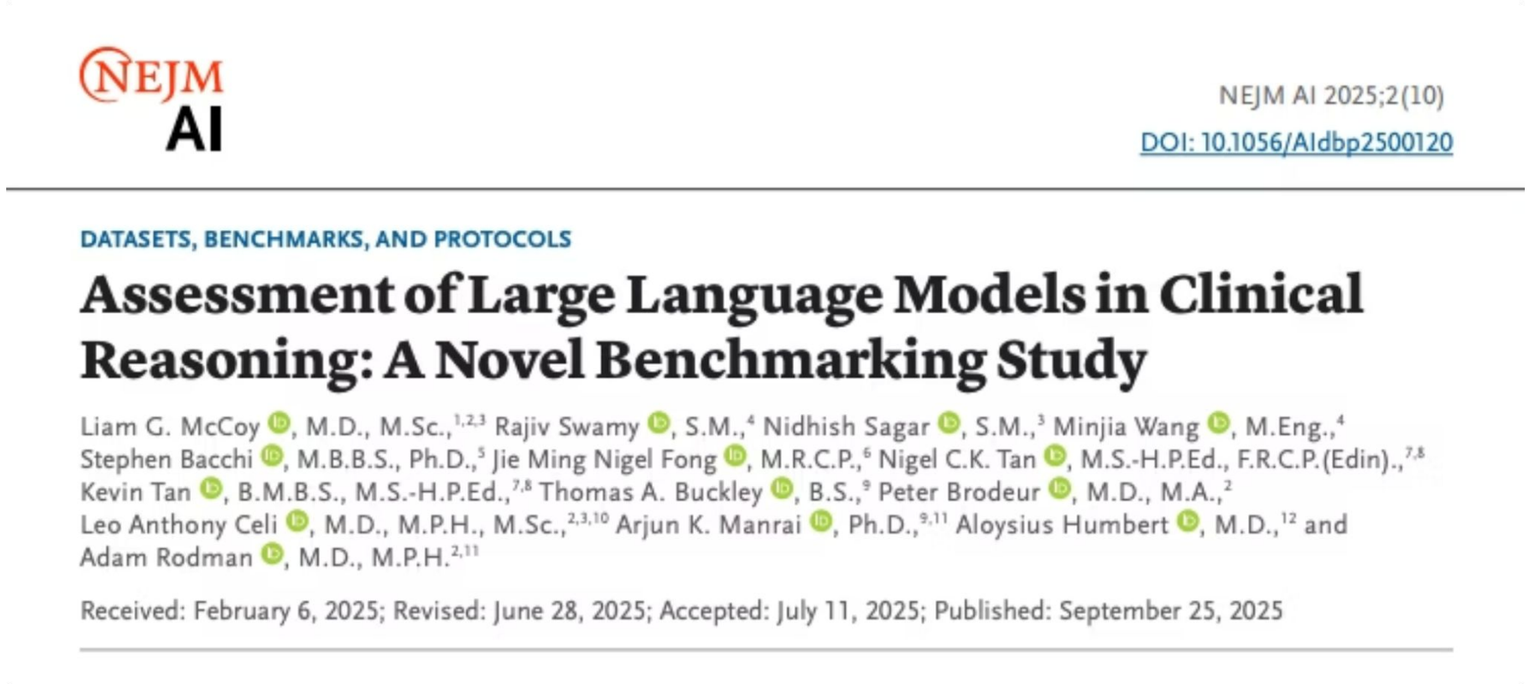
→ Benchmarks, lab datasets

Effectiveness

Does it work in the **real world**?

→ Clinical deployment, prospective trials

⊗ **Critical gap:** Most AI studies assess efficacy only. We cannot yet infer real-world effectiveness from benchmark performance alone.



NEJM AI

NEJM AI 2025;2(10)
DOI: 10.1056/AIdbp2500120

DATASETS, BENCHMARKS, AND PROTOCOLS

Assessment of Large Language Models in Clinical Reasoning: A Novel Benchmarking Study

Liam G. McCoy, M.D., M.Sc.,[1,2,3] Rajiv Swamy, S.M.,[4] Nidhish Sagar, S.M.,[3] Minjia Wang, M.Eng.,[4] Stephen Bacchi, M.B.B.S., Ph.D.,[5] Jie Ming Nigel Fong, M.R.C.P.,[6] Nigel C.K. Tan, M.S.-H.P.Ed., F.R.C.P.(Edin).,[7,8] Kevin Tan, B.M.B.S., M.S.-H.P.Ed.,[7,8] Thomas A. Buckley, B.S.,[9] Peter Brodeur, M.D., M.A.,[2] Leo Anthony Celi, M.D., M.P.H., M.Sc.,[2,3,10] Arjun K. Manrai, Ph.D.,[9,11] Aloysius Humbert, M.D.,[12] and Adam Rodman, M.D., M.P.H.[2,11]

Received: February 6, 2025; Revised: June 28, 2025; Accepted: July 11, 2025; Published: September 25, 2025

Critical Appraisal — Benchmarking

NEJM AI 2025;2(10) · DOI: 10.1056/AIdbp2500120

Study Type

Datasets, Benchmarks & Protocols

Focus

LLM clinical reasoning via Script Concordance Testing

Published

September 25, 2025

How To Read An AI Paper In Under 15 Minutes: A PEM Pocket Framework

1. What clinical task is the model doing?

If you cannot name the task in one sentence, the paper is not ready to translate to your shop.

2. Efficacy or effectiveness?

If efficacy only, you cannot conclude the tool will help your patients - only that it could.

3. What was the comparator?

Many AI papers benchmark against the weakest plausible human comparator and then headline "AI matches doctors." Read the comparator carefully.

4. What was the prompt or benchmark?

Was the model spoon-fed a clean, complete vignette - or did it have to gather information itself, the way you do at the bedside? Performance usually drops sharply when the model has to ask for the data. The benchmark world is not the ED.

5. Who's missing from the data?

Pediatric patients are often invisible in adult-trained models, and tools that look great in adults can fail silently on neonates and infants.

6. What kinds of errors does the model make, and are they the same errors a clinician would make?

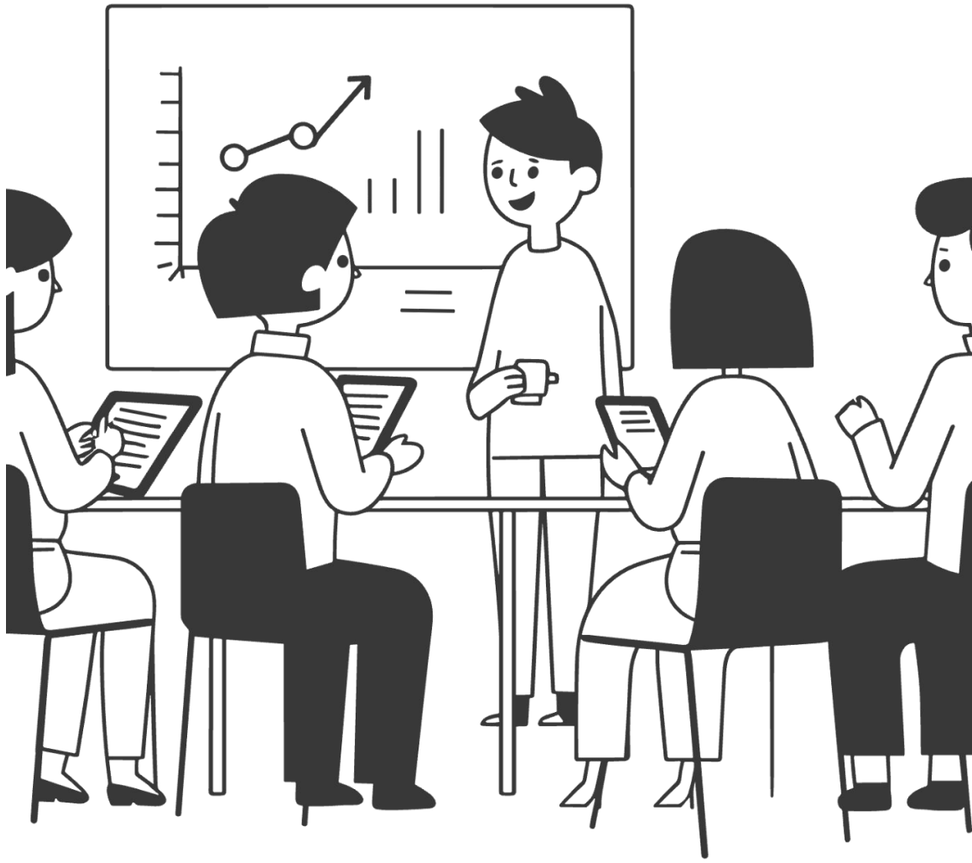
A model that confidently produces extreme errors with no way to know it is wrong is the worst case.

Bonus question for vendor pitches

If a vendor walks into your group meeting next month with this tool, what specific evidence would you need to see before letting them deploy on your patients?

Session Outline

1. **Abstract**
   5 min — Read & Discuss
2. **Introduction**
   5 min — Discuss
3. **Methods**
   15 min — Read & Discuss
4. **Results / Conclusions**
   15 min — Read & Discuss



Abstract : Key Questions

Aims of the Study

Provide an open-source Script Concordance Test (SCT) dataset for benchmarking clinical reasoning; compare LLM vs. human performance on SCTs.

Key Results

LLMs perform **worse on SCTs than on other benchmarks** (e.g., board MCQs), and generally **underperform most human comparators** — except medical students.

Abstract — Critical Lens

1

Novel Benchmark

Open-source SCT dataset addresses a key gap in clinical reasoning evaluation

2

SCT vs. MCQ

SCTs capture reasoning under uncertainty; MCQs may reflect memorized knowledge

3

Human Comparators

LLMs outperform only medical students — do worse compared to residents, attendings

4

Open Science

Dataset publicly released - enables reproducible, standardized future benchmarking



Session Outline

1. Abstract
   ✓ Complete
2. **Introduction**
   5 min — Now Discussing
3. Methods
   10 min — Up Next
4. Results / Conclusions
   10 min — Coming Soon

Introduction — Context

Large language models (LLMs) have been rapidly integrated into everyday clinical care, transcribing encounters,[1-3] drafting patient messages, summarizing clinical documentation,[4,5] and providing point-of-care clinical decision support.[6,7] With workforce shortages accelerating interest in LLM-augmented reasoning, responsible deployment demands robust, standardized evaluation. Although benchmarks cannot replace clinical trials, they have become indispensable early filters.

Driving Forces

- LLMs rapidly integrated into clinical care.
- Workforce shortages accelerating interest in LLM-augmented reasoning.
- Responsible deployment requires robust, standardized evaluation.
- Benchmarks: indispensable early filters.

Introduction — Knowledge Gap

Benchmarks for Clinical Reasoning

MCQs / Board Questions

Likely measures *encoded knowledge* rather than clinical reasoning

Open-ended Clinical Cases

e.g., NEJM CPC series — high face validity, hard to standardize

Med-Ed Measures (OSCE)

High validity, but resource-intensive to score

Figure 3. Quality of o1-preview Diagnostic Test Selection on NEJM Clinicopathologic Case Conferences (CPCs)

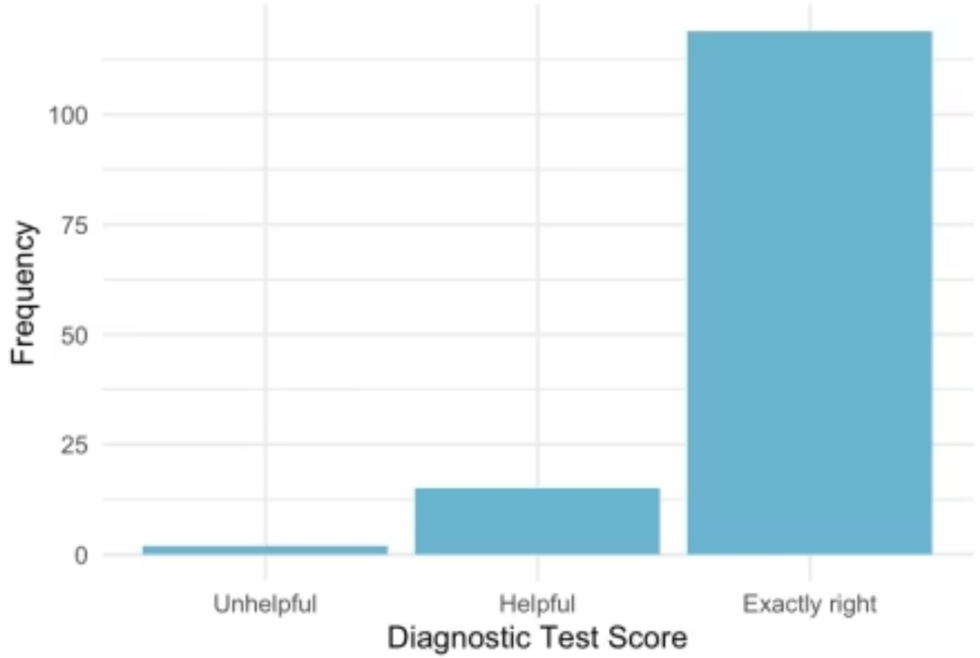
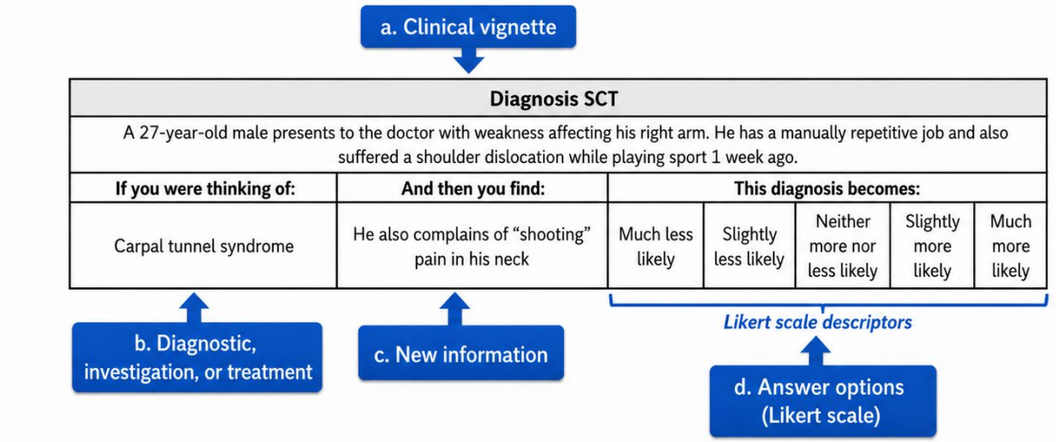


Figure 3: Performance of o1-preview in predicting the next diagnostic tests that should be ordered. Performance was measured by two physicians using a likert scale of "Unhelpful," "Helpful," and "Exactly right." We excluded 7 cases from the total case set in which it did not make sense to ask for the next test (Supplement 1B).

Introduction — Script Concordance Testing

Discussion prompt: Do you think SCTs make a good benchmark for clinical reasoning? Why or why not?



Diagnosis SCT							
A 27-year-old male presents to the doctor with weakness affecting his right arm. He has a manually repetitive job and also suffered a shoulder dislocation while playing sport 1 week ago.							
If you were thinking of:	**And then you find:**	**This diagnosis becomes:**					
Carpal tunnel syndrome	He also complains of "shooting" pain in his neck	Much less likely	Slightly less likely	Neither more nor less likely	Slightly more likely	Much more likely	

Clinical Stem	If you were thinking of	And then you find:	Category	−2 Score	−1 Score	0 Score	1 Score	2 Score
A 27-year-old male presents to the doctor with weakness affecting his right arm. He has a manually repetitive job and also suffered a shoulder dislocation while playing sport 1 week ago.	Carpal tunnel syndrome	He also complains of "shooting" pain in his neck	Expert opinions	10	7	0	0	0
			Score values	1.0	0.7	0	0	0

Scores are derived from the distribution of expert responses — weighted by expert consensus, not ground truth.

SCT: *Limitations*

1. They are usually **single-turn** questions.
2. They do not test whether the model can **ask the right questions**.
3. They do not test whether it can gather missing data.
4. They do not fully test management, reassessment, uncertainty, or safety.
5. They rely on expert consensus, which may not always equal truth.
6. They are still mostly **efficacy benchmarks**, not testing effectiveness.

Session Outline

1 — Abstract

✓ Complete

2 — Introduction

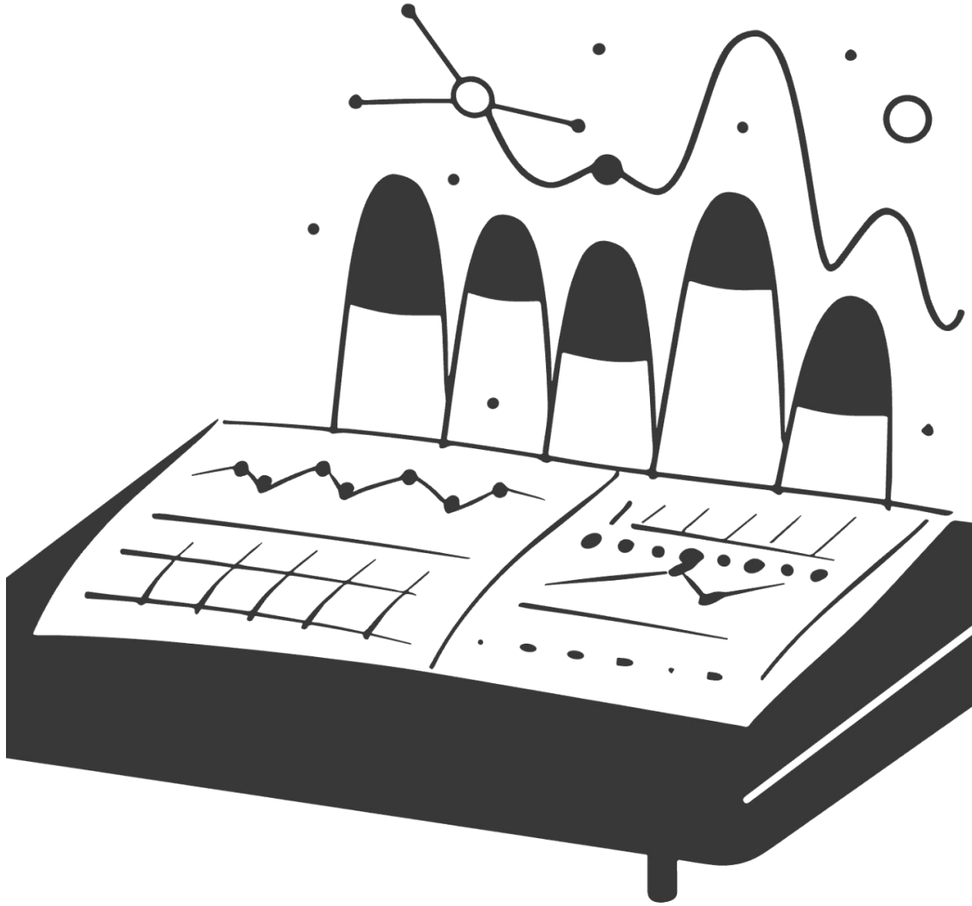
✓ Complete

3 — **Methods**

15 min — Now Discussing

4 — Results / Conclusions

15 min — Coming Soon



Methods — Models Evaluated

10 LLMs tested with **default settings**

OpenAI

- GPT-4o
- o1-preview
- o4-mini
- o3

Anthropic

- Claude 3.5 Sonnet

Google

- Gemini 1.5 Pro
- Gemini 2.5

Meta / Mistral

- Llama 3.3 70B
- Mistral Small 24B

Methods — Model Deep Dive: o3

In **3 minutes**, search the internet for information about the **o3** model and answer the following:

1. Type of AI model?
2. Developer?
3. Foundation, reasoning, or agent?
4. Training data — type, sources, time range?
5. Training techniques used?
6. Big model or small/efficient model?

Methods — o3 Model: Answers



Type

Generative AI / Large Language Model

Developer

OpenAI

Model Class

Reasoning model with some agentic capacity

Training Data

Multimodal (text + images); internet sources (Wikipedia, GitHub, etc.) + synthetic data; likely through 2024–2025. **Limited transparency.**

Training Technique

Reinforcement learning

Size

Large frontier model — many features, not optimized purely for efficiency

Methods — Prompts

> Which **prompt engineering strategies** were used in the study?

Zero-shot Prompting

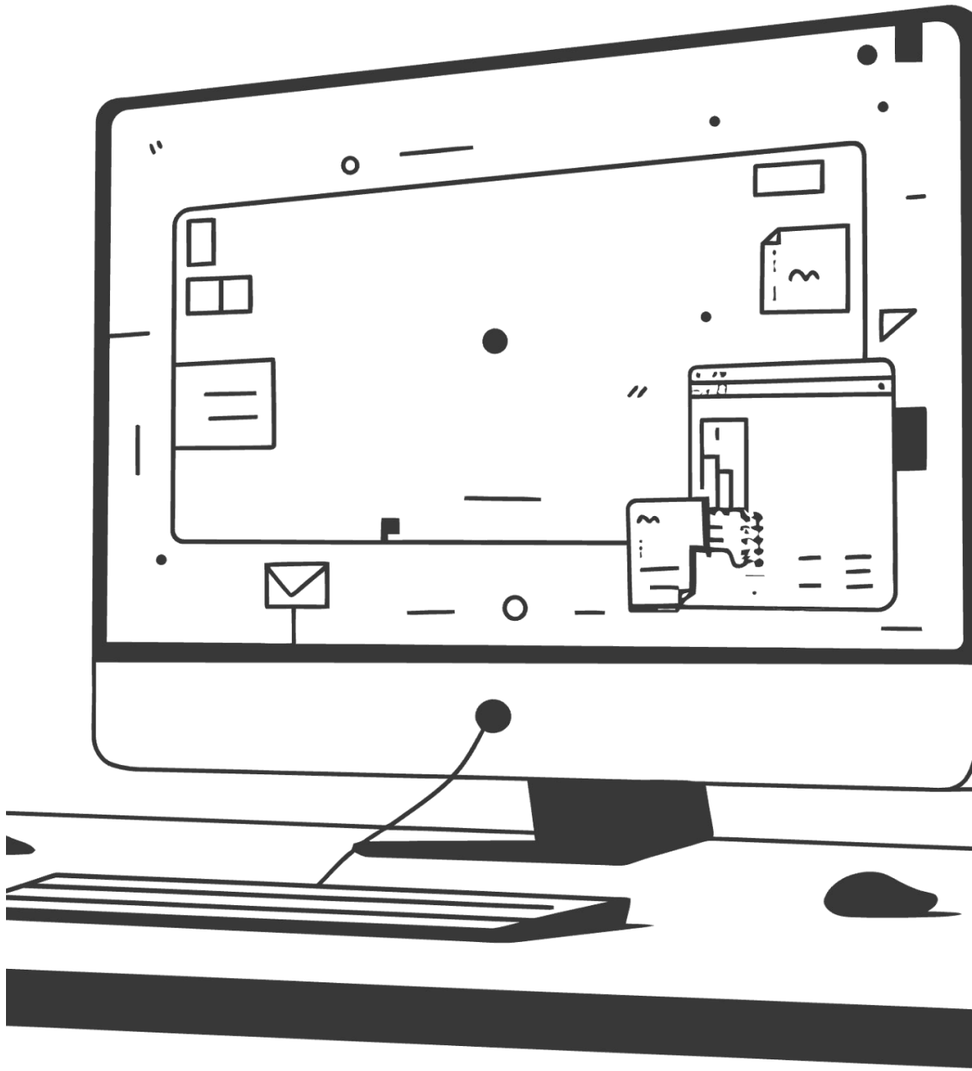
No examples provided — model answers from instructions alone

Few-shot Prompting

A small number of worked examples included in the prompt

Chain-of-Thought

Model prompted to reason step-by-step before answering



Basic Instructions in Prompt

The prompt framed the task as a **Script Concordance Test (SCT)** — evaluating how new information shifts the likelihood of a clinical hypothesis.

Script Concordance Testing
You are taking a Script Concordance Test, which evaluates your understanding of medical knowledge.In this test, you will evaluate how new information impacts a specific hypothesis. Use the following scale to rate the impact:
-2: Strongly decreases the likelihood of the hypothesis -1: Slightly decreases the likelihood of the hypothesis 0: No effect on the likelihood of the hypothesis +1: Slightly increases the likelihood of the hypothesis +2: Strongly increases the likelihood of the hypothesis

Response format
Respond with "Rating: " followed by a rating (-2, -1, 0, +1, +2) and a brief explanation for your choice. This is an exam, and you are required to provide a valid answer.

Rating Scale

−2 (strongly decreases) to +2 (strongly increases) likelihood

Response Format

Rating + brief explanation required; exam conditions enforced

Few-Shot Example (5 per prompt)

Each example provided a **clinical scenario**, a hypothesis under consideration, new information, and a rated response with explanation.

Scenario
A 2-year-old female has a 2-day history of left ear pain. Her mother has been giving her acetaminophen with temporary improvement in pain. You elicit further history from the mother. The mother reports that the patient has a history of multiple previous episodes of acute otitis media that have required oral antibiotics. You complete a physical exam on the patient. The patient's mother inquires about your management plans.

Hypothesis
If you were thinking of: Prescribing oral amoxicillin for 10 days

Additional Information
And then you find: An erythematous, swollen external ear canal with white discharge

Response
Rating: -2

Explanation:
The additional information indicates the presence of an erythematous, swollen external ear canal with white discharge, which is consistent with otitis externa (commonly known as swimmer's ear). Otitis externa is typically caused by bacterial infections such as Pseudomonas aeruginosa or Staphylococcus aureus and is treated with topical antibiotics rather than systemic antibiotics like oral amoxicillin. Prescribing oral amoxicillin for 10 days would be inappropriate and ineffective for treating otitis externa. This strong contraindication to the initial hypothesis of prescribing oral amoxicillin significantly decreases the likelihood of that management plan being appropriate.

Example: 2-year-old with ear pain → hypothesis: oral amoxicillin for 10 days → new finding: otitis externa → Rating: −2 (strong contraindication)

Examples anchor the model to the expected reasoning format and calibrate response quality before the test item.



Session Outline

01

Abstract

Read and discuss — 5 min

02

Introduction

Discuss — 5 min

03

Methods

Read and discuss — 15 min

04

Results / Conclusions

Read and discuss — 15 min



DISCUSSION PROMPT

Results:

Does this paper evaluate effectiveness or efficacy?

KEY DISTINCTION

Efficacy - Not Effectiveness

This Study

Controlled experiment : not an implementation study that shows how outcomes are changed.

What's the difference?

Efficacy

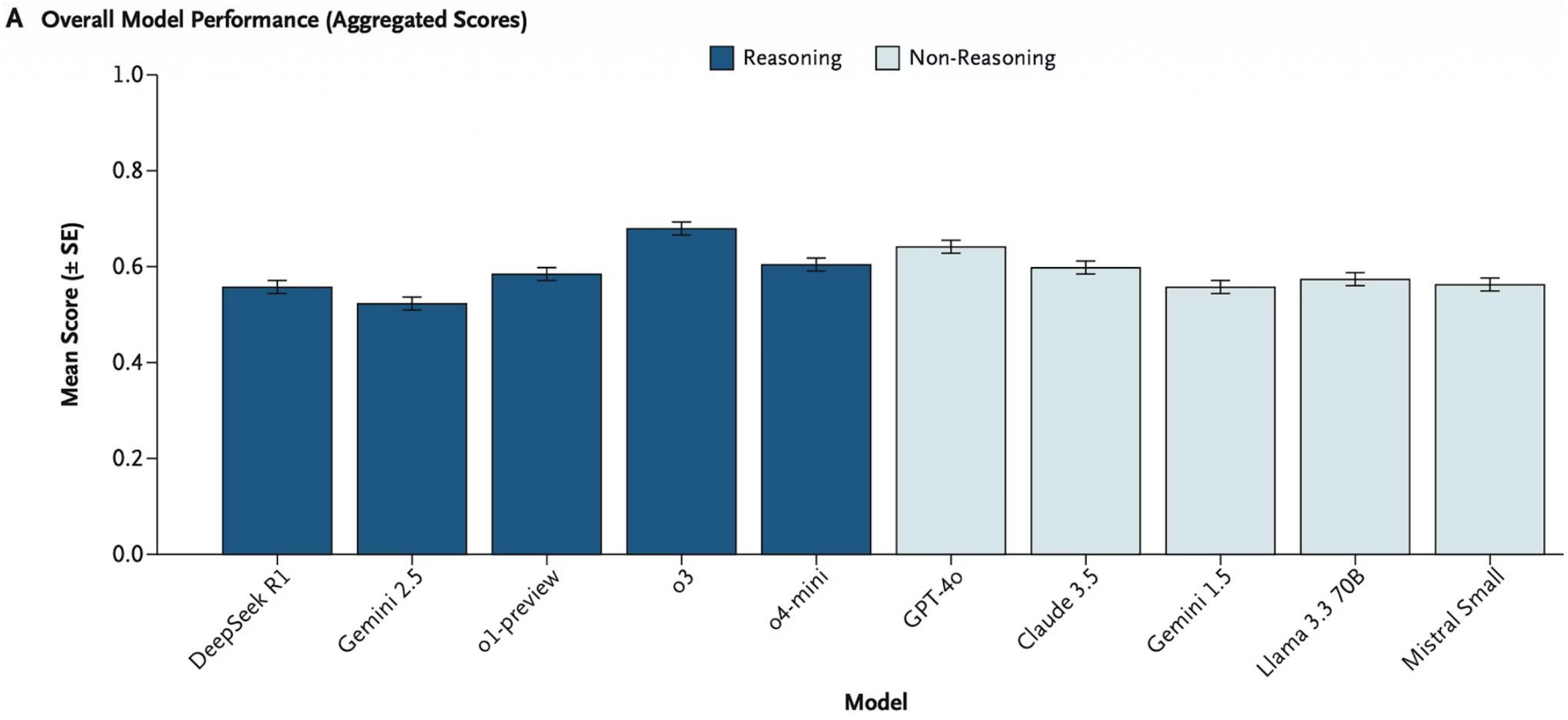
Performance under controlled, ideal conditions

Effectiveness

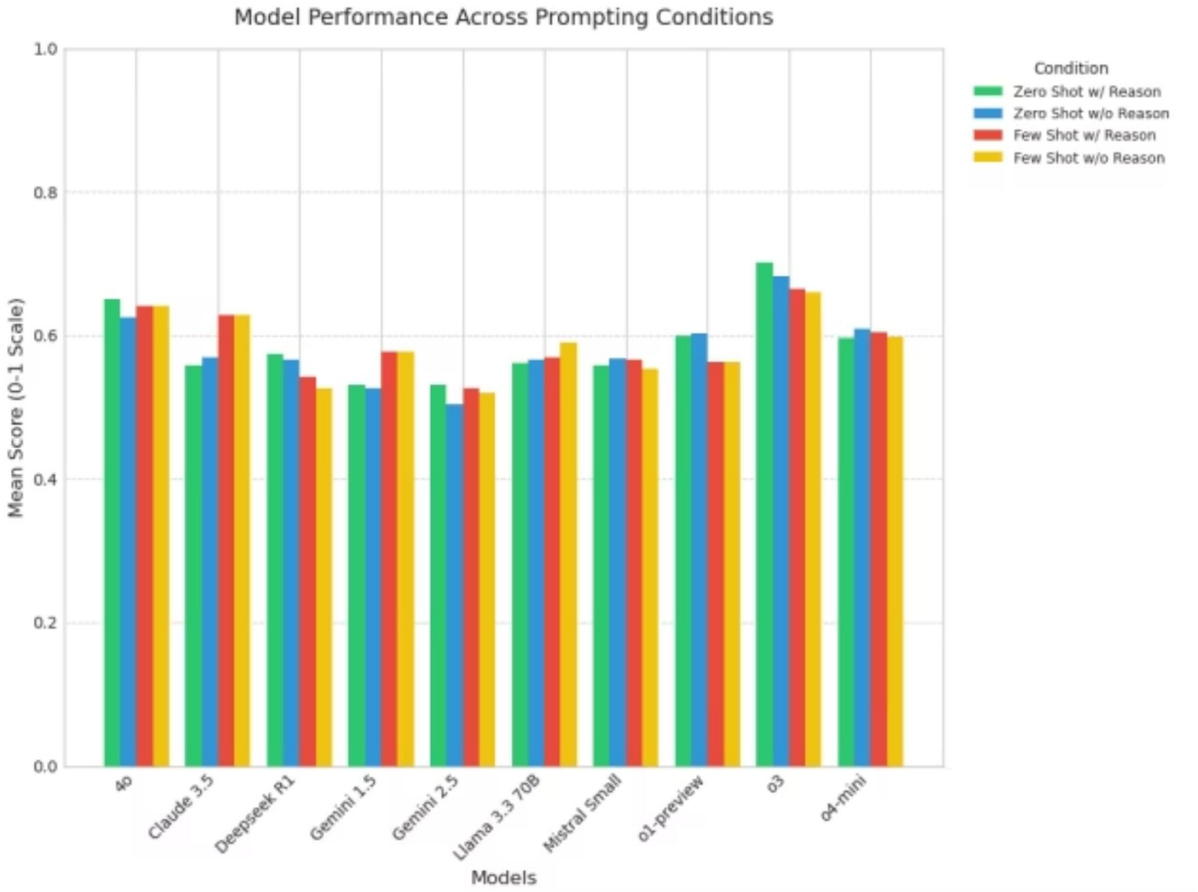
Performance in real-world clinical settings

Model vs. Model Performance (Fig 2a)

Aggregated SCT scores across all prompting conditions. Reasoning vs. non-reasoning models compared.



Model Performance — Key Finding (Fig S2)



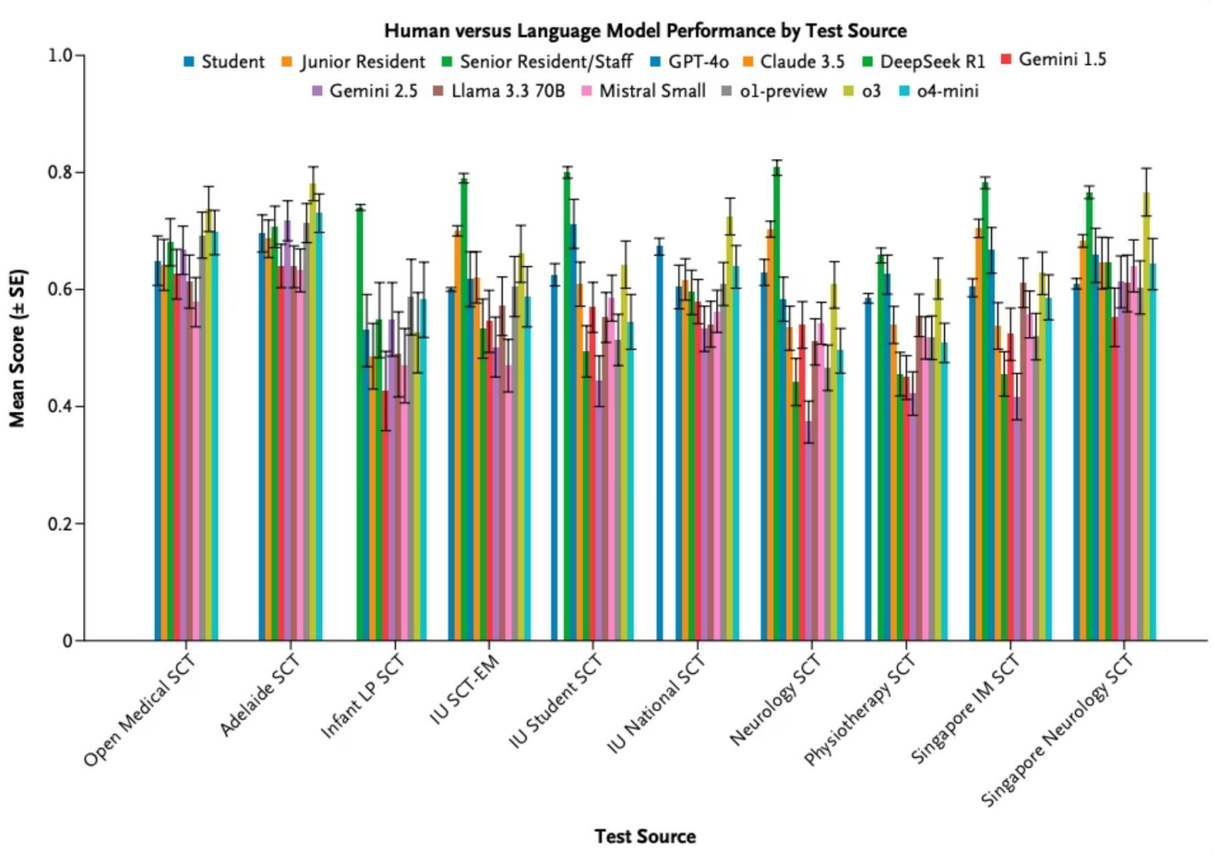
o3 edged ahead — but the difference was minimal

No meaningful advantage found for:

- Reasoning vs. foundation models
- Zero-shot vs. few-shot prompting
- Chain-of-thought vs. no chain-of-thought

All models clustered in a similar performance range (~0.5–0.7).

Model vs. Human — Key Finding (Fig 3)



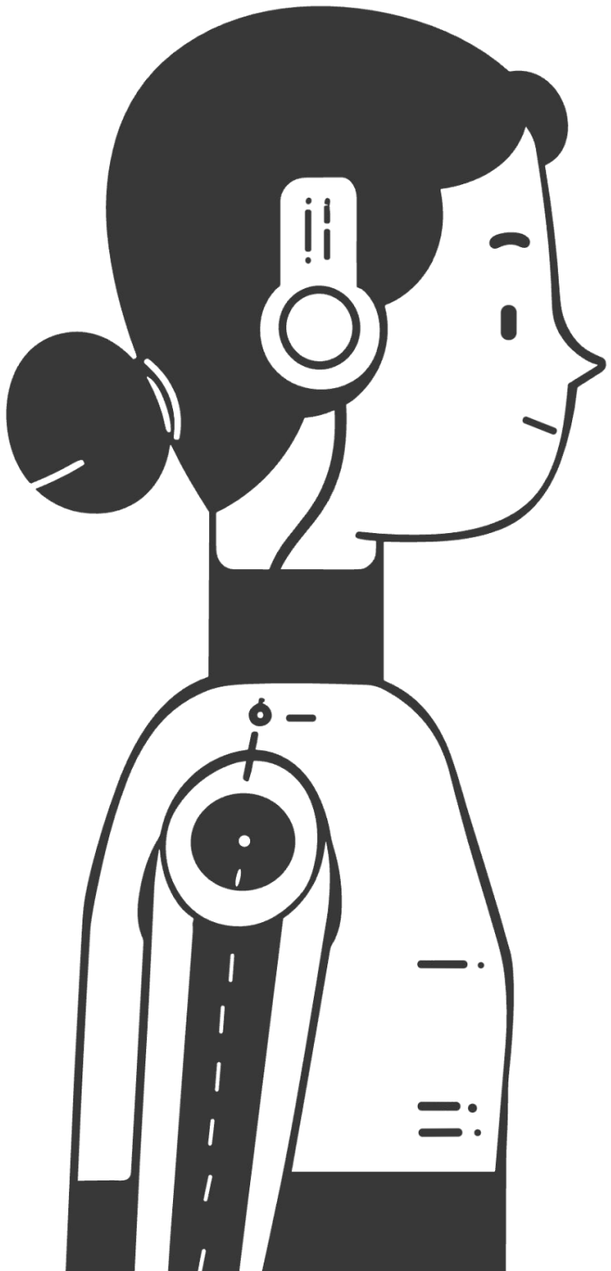
vs. Students

Models ≈ comparable performance

vs. Residents / Attendings

Senior clinicians outperformed all models

⚠ Clinical expertise still surpasses LLMs on SCT benchmarks



Reasoning Patterns - Key Takeaway

More Extreme Answers

Models chose –2 or +2 more often; humans favored moderate ratings

Average ≠ Accurate

Models scored well on average but got fewer questions *absolutely* correct

Different Errors

Models and humans failed on different questions - distinct reasoning pathways

Paper Summary

SCTs as Benchmarks

Potentially more valid for clinical reasoning than MCQs

Models ≈ Models

No meaningful differences between models or prompt strategies

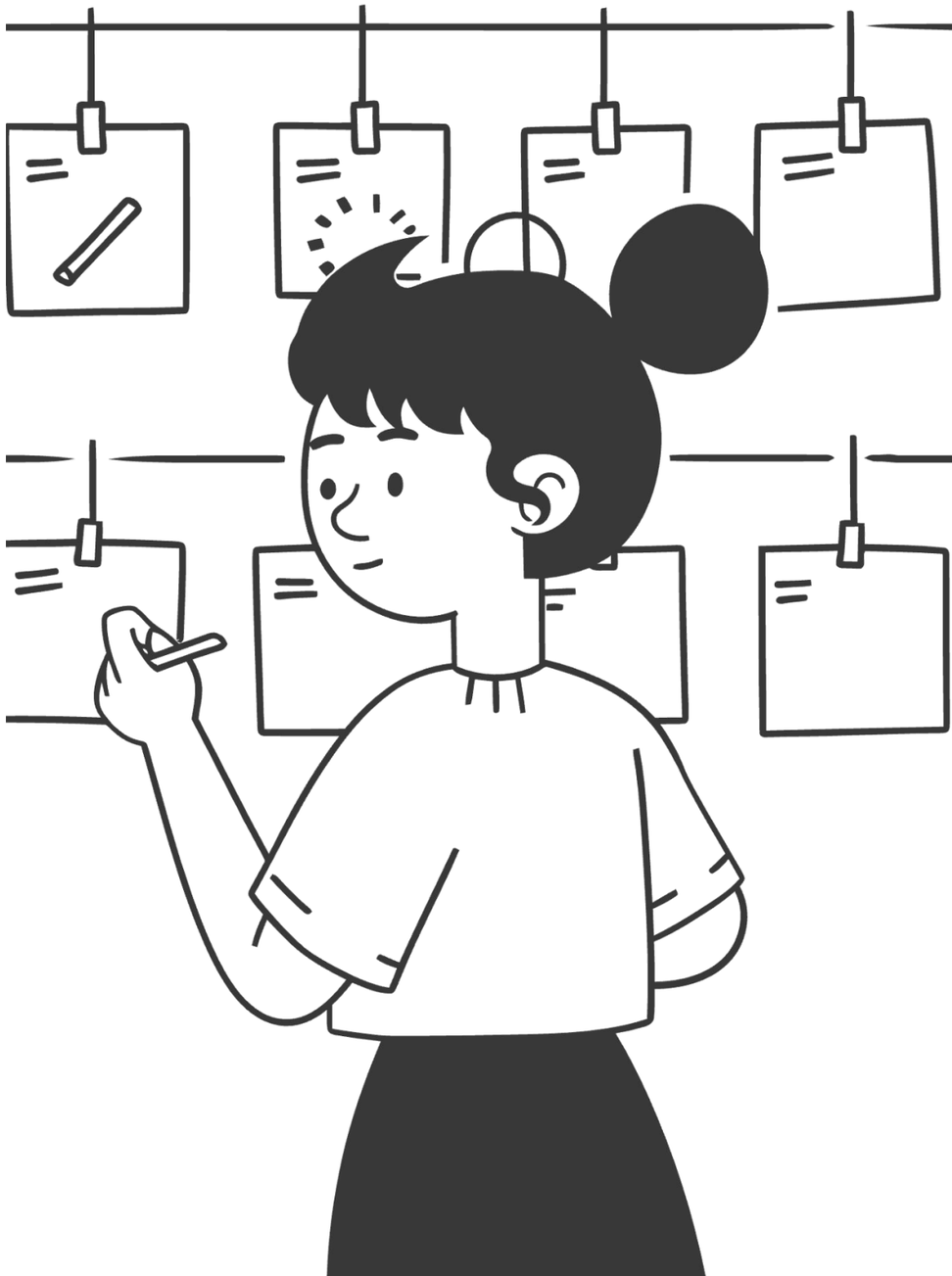
Models vs. Humans

Outperformed students; fell short of residents and faculty

Different Reasoning

Models showed a fundamentally different pattern of responses than humans

⚠ ⚠ No RCTs of LLMs used for clinical reasoning in a real clinical context





Course Outline

01

Introduction to AI

What is AI? Why do we care?

02

Error and Bias in AI Systems

AI for Clinical Decision-Making

03

Evaluating and Appraising AI

AI Benchmarks; Critical Appraisal

04

Critical Appraisal + Clinical Applications of AI

Documentation; Future Directions